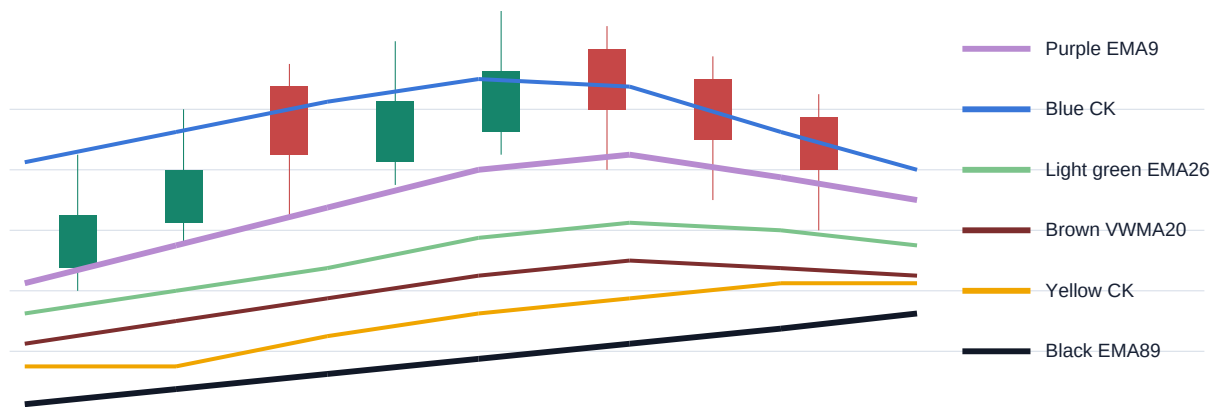


PURPLE TOUCH

Implementation and Review Guide

A visual explanation of how the scanner shortlists F&O stocks, applies the confirmed Krishna rules, waits for entry confirmation, creates alerts and trade IDs, and tracks exits.



Concept sketch only - values are illustrative, not a market chart.

Purpose

Use this document to compare the implemented logic with Krishna's intended chart setup. It documents current code behavior and calls out assumptions that may need refinement.

Scope and safety

The feature identifies analysis candidates and tracks confirmation/invalidation context. It does not place orders, execute trades, or claim future performance.

Version 1.2 | Branch scanner-audit-and-v3 | Generated 30 Aug 2026

1 The complete workflow at a glance

The system separates discovery, qualification, entry, and exit. A stock appearing in Step 1 is not automatically an alert. A strict candidate is also not automatically a new alert.

1. Refresh cached candles

Download/update all required Monthly, Weekly, Daily and lower-timeframe candles for the F&O list.

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2. Step 1 - purple EMA9 distance

Latest higher-timeframe candle must touch EMA9 or come within 3.00%. Rows rank from 0.00% upward.

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3. Strict higher-timeframe qualification

Apply EMA89, EMA26, blue-above-purple, blue-to-purple approach, higher confirmation, and structure checks.

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4. Entry status

Evaluate early and final entry timeframes. Status can be waiting, candidate, discarded, missing, or insufficient.

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5. Alert + trade ID

During NSE market hours, a fresh entry candidate creates one entry event and one open trade ID. Duplicates are suppressed.

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6. Exit tracking

Only open trade IDs are checked. A close below yellow on the configured exit timeframe creates an exit alert and closes that ID.

Core distinction

Step 1 rows > strict candidates > entry-ready rows > newly created alerts. Each stage can only reduce or preserve the previous stage count.

2 Timeframe matrix

All three profiles are scanned independently because an opportunity can originate from Monthly, Weekly, or Daily purple touch.

Profile	Touch timeframe	Higher confirmation	Early entry	Final entry	Exit
Monthly	Monthly	Derived 5-month candle: open or close above blue	120 minute (2 hour)	Daily	120 minute
Weekly	Weekly	Monthly candle: open or close above blue	30 minute	120 minute (2 hour)	30 minute
Daily	Daily	Weekly candle: open or close above blue	10 minute	30 minute	10 minute

Independent monitor intervals

Profile	UI interval choices	What the interval controls
Monthly	15m, 30m, 1h, 2h (default)	How often the Monthly profile refresh/check is queued during market hours.
Weekly	5m, 10m, 15m, 30m (default)	How often the Weekly profile refresh/check is queued during market hours.
Daily	3m (default), 5m, 10m	How often the Daily profile refresh/check is queued during market hours.

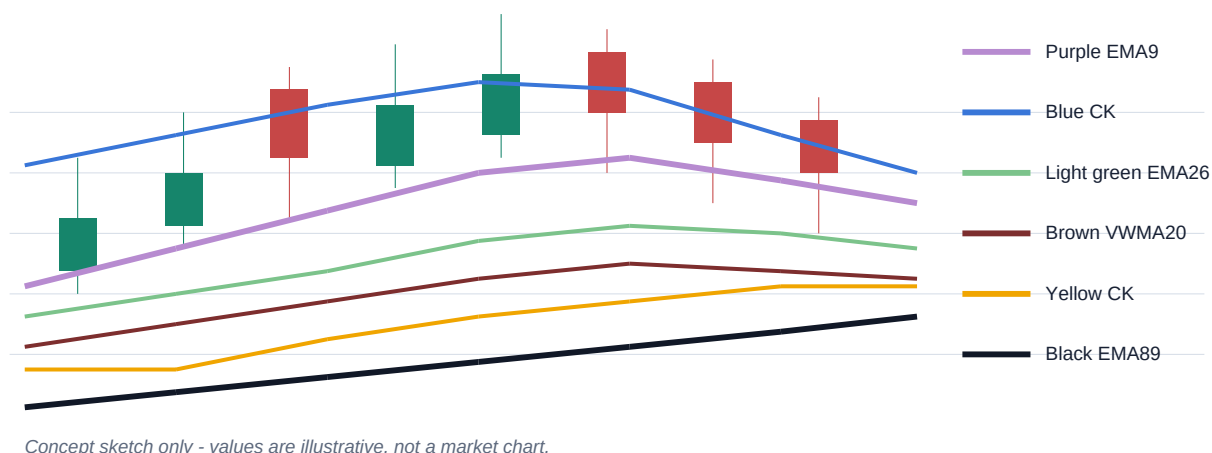
No overlapping scan work

Due profile scans are queued. Candle refresh, entry checks, and exit checks for the same run complete in sequence so concurrent scans do not overwrite UI state.

Per-profile visibility

The UI shows separate Monthly, Weekly, and Daily progress bars plus each profile's last start, last end, and next scheduled trigger. A queued profile waits for the active profile to finish.

3 Indicator and color mapping



Color	Implemented indicator	Role in this setup	Mandatory?
Purple	EMA 9	Touch line and fast trend reference.	Yes - range distance <= 3.00%
Blue	Upper Chande Kroll stop (10, 1, 9)	Must be above purple; used for approach and higher confirmation.	Yes
Yellow	Lower Chande Kroll stop (10, 1, 9)	Entry close-above trigger and exit close-below trigger.	Yes
Light green	EMA 26	Higher-timeframe trend filter; close and EMA9 relationships are displayed.	Yes in strict scan
Brown	VWMA 20	Final entry requires yellow above brown; early entry treats it as context.	Final only
Black	EMA 89	Longer trend filter; higher-timeframe close must be above it.	Yes
RSI	RSI 14	Bullish divergence can improve context but does not block entry.	Optional

Explicitly ignored for Purple Touch

Ichimoku cloud, VWAP, and Donchian Channel 20 may exist elsewhere in the project but are not qualification conditions in this setup.

4 Step 1 - Purple EMA9 touch and distance

The latest candle on the selected touch timeframe must physically contain EMA9 within its high/low range or its nearest edge must be no more than 3.00% away. Rows are ranked from the smallest range distance upward.



Implemented expression

PASS when EMA9 is inside the candle range (0.00%) or the nearest candle edge is within 3.00% of EMA9. Close distance remains separate context. UI bands show exact 0.00%, then 0.20%, 0.50%, 1.00%, 1.50%, 2.00%, and the full 3.00% shortlist.

What Step 1 does and does not mean

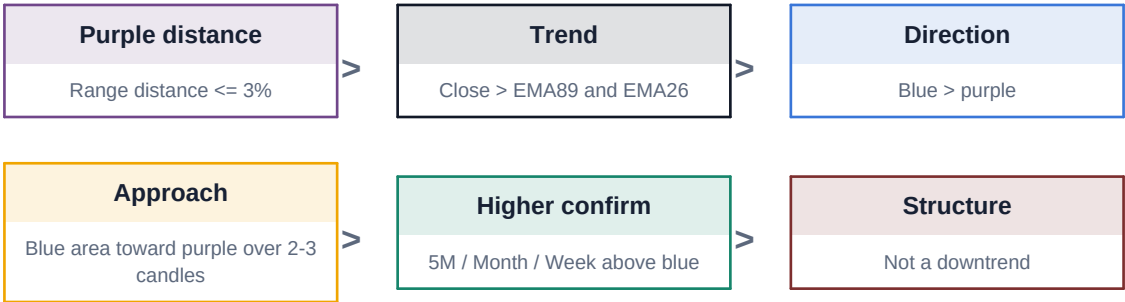
Step 1 result	Meaning	Next action
Touch row	The latest Monthly/Weekly/Daily candle range touches EMA9 or comes within 3.00%.	Audit the strict blockers shown in the same row.
No row	EMA9 is more than 3.00% from the latest candle range, or required data is unavailable.	The stock does not proceed in that profile for this scan.
Same symbol in multiple rows	The symbol independently touched in more than one profile.	Review each profile using its own entry/exit matrix.

Confirmed distance rule

The current maximum is 3.00%. Near-touch rows may proceed through strict qualification and entry alerts. UI filters narrow the view without changing the scanner result set.

5 Strict higher-timeframe qualification

A Step 1 touch becomes a strict candidate only when every mandatory higher-timeframe gate passes.



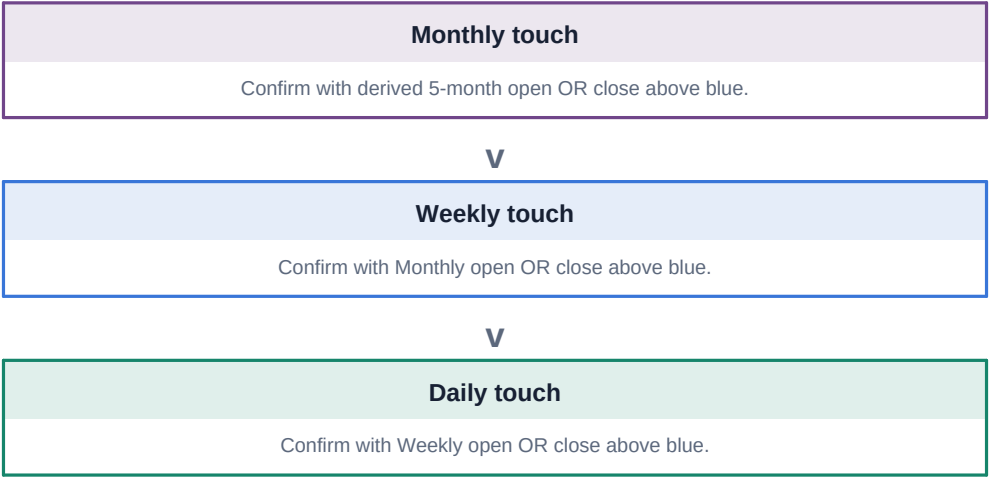
Gate	Pass condition	UI result when it fails
Black EMA89	Latest higher-timeframe close > EMA89.	Block; Unknown when fewer than 89 candles are available.
EMA26	Latest close >= EMA26. EMA9 > EMA26 adds constructive context.	Block when close is below EMA26.
Blue / purple	Upper Chande Kroll blue line > purple EMA9.	Block when blue is not above purple or cannot be calculated.
Approach	Within latest 3 candles, a prior close was near blue and latest price moved toward purple.	Block when direction is not observed.
Higher confirmation	Profile-specific confirmation candle passes blue-line rule.	Block or Missing/Insufficient.
Market structure	Structure is uptrend/range/neutral; explicit downtrend is rejected.	Block when downtrend.

Strict candidate is not an alert
The strict table can include rows whose early and final entry status is still 'wait'. Alert creation only evaluates rows with entry_candidate status.

6 Higher confirmation by profile

The blue-line confirmation is evaluated on a separate higher aggregation. This is mandatory in the current implementation.

Purple touch	Confirmation data	PASS	BLOCK
Monthly	Daily candles are aggregated into stable calendar 5-month buckets.	Latest derived 5-month open OR close > its blue Chande Kroll.	Both open and close are <= blue, or blue/data is unavailable.
Weekly	Monthly candles.	Latest Monthly open OR close > its blue Chande Kroll.	Both Monthly open and close are <= blue, or blue/data is unavailable.
Daily	Weekly candles.	Latest Weekly open OR close > its blue Chande Kroll.	Both Weekly open and close are <= blue, or blue/data is unavailable.

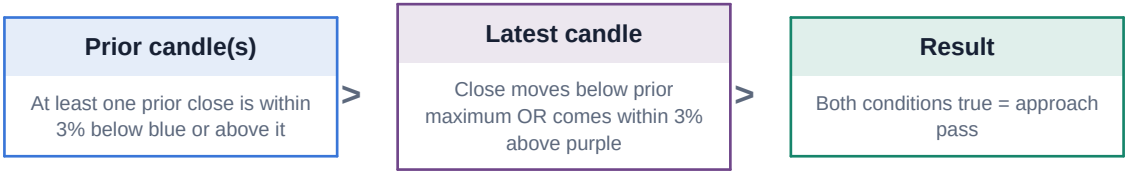


Review point for Krishna

The 5-month candle is derived by grouping source candles into stable calendar buckets. Confirm whether Krishna intended fixed Jan-May / Jun-Oct style buckets, a rolling 5-month candle, or the broker chart's own aggregation.

7 Blue-to-purple approach

This gate attempts to encode Krishna's observation that blue should be above purple and the candle should come from the blue area toward purple. It is the least visually exact part of the current rules and deserves deliberate review.



Current parameters

Parameter	Current value	Why it exists	Potential improvement
Lookback	Latest 3 touch-timeframe candles	Supports the accepted 'last 2-3 candles' guidance.	Allow 2 or 3 as a UI/config choice.
Blue proximity	3%	Allows a candle to be treated as coming from the blue area.	Replace percentage with ATR or candle-range distance.
Movement toward purple	Latest close declines from prior max OR is within 3% above EMA9	Captures pullback toward purple.	Use distance-to-purple that must contract candle by candle.

Important

Both Step 1 distance and the directional approach currently use a 3% ceiling, but they measure different things: Step 1 measures candle-range distance to EMA9; approach measures movement from blue toward purple.

8 Entry lifecycle - Candle 1, Candle 2, Candle 3+

After a valid purple touch, the setup remains pending until lower-timeframe candles establish a reference and confirmation can be evaluated.



Stage	Implemented behavior	Status
Candle 1	Map the higher-timeframe touch period onto the entry timeframe. Record Candle 1 low.	wait
Candle 2	Wait for the next entry-timeframe candle to close. Record Candle 2 low.	wait until closed
Candle 3 onward	Entry rules may qualify. Continuously check whether any later candle breaks the lower of Candle 1 and Candle 2 lows.	entry_candidate or wait
Invalidation	Any later candle low < min(Candle 1 low, Candle 2 low).	discarded
Expiry	No fixed 10-candle expiry is used.	remains active until discarded or entered

Implemented invalidation level
The UI reports min(Candle 1 low, Candle 2 low) as the effective threshold. Breaking only the higher reference low does not discard the setup. The first later candle below the lower reference low records the invalidation timestamp.

9 Early entry versus final entry

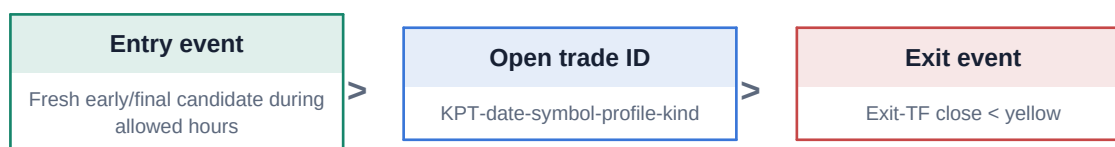
Both entry styles use the profile matrix, but final entry has one additional mandatory relationship.

Rule	Early entry	Final entry
Entry timeframe	Monthly 2H; Weekly 30m; Daily 10m	Monthly Daily; Weekly 2H; Daily 30m
Candle 2 closed	Mandatory	Mandatory
No C1/C2 low break	Mandatory	Mandatory
Entry close above yellow	Mandatory	Mandatory
Blue above purple on entry timeframe	Mandatory	Mandatory
Yellow above brown VWMA20	Optional context	Mandatory
RSI bullish divergence	Optional quality context	Optional quality context
Close above blue	Rare stronger context	Rare stronger context



Why a strict candidate can still show Wait
Higher-timeframe qualification and lower-timeframe entry confirmation are separate. A stock can pass every strict filter while its lower-timeframe close remains below yellow or Candle 2 has not closed.

10 Alert, trade ID, and exit lifecycle



Item	Meaning	Persistence behavior
Entry Alert History	All historical entry events created by the scanner.	Rows remain after exit; history is not deleted.
Open Trade IDs	Entry events that have not yet received an exit event.	Only these IDs are checked by the exit scan.
Exit Alerts	Historical exit events linked to trade IDs.	Closes the matching open ID.
Duplicate suppression	Same symbol + touch profile + early/final kind while already open.	Does not create another trade ID on each monitor pass.

Count reconciliation

Historical entry events = open trade IDs + closed trade IDs. Closed trade IDs should equal exit alerts. Example: 91 entries, 91 open, 0 exits is internally consistent.

Exit rule

For both early and final entries, the scanner loads the configured exit timeframe and checks the latest candle. An exit candidate is generated only when that candle closes below the yellow Chande Kroll line. It does not scan unrelated symbols for exits.

11 What happens when Run scanner now is clicked

Refresh queued

The bulk job count is unique market targets multiplied by source timeframes. 2-hour analysis uses downloaded 1-hour source candles.

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Refresh completed

Completed/total means broker/cache refresh ended. It is not the end of setup analysis.

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Analysis running

The service scans all selected symbols and profiles, calculates indicators, applies gates, and checks open exits.

V

Analysis completed

UI renders Step 1, strict candidates, early/final-ready counts, and newly created alert counts.

Control/state	Behavior
Run scanner now disabled	A refresh or analysis request is still in flight. It re-enables only when both phases finish or fail.
Cancel current scan	Stops the bulk refresh between broker requests. It cannot interrupt an already executing broker request or the later pure analysis phase.
Start market monitor	Starts recurring market-hour profile scans at independent Monthly/Weekly/Daily intervals.
Force cached-data outside market hours	Off: analyze only. On: also permits test entry/exit trade IDs and optional Telegram messages from cached candles.

Normal setting

Keep Force off. Outside NSE market hours, candidate tables can refresh but new alert events, trade IDs, exit events, and Telegram messages are intentionally paused.

12 How to read the UI counts

UI count	Definition	Does it create an alert?
Analyzed	Stock-profile rows with enough source data to run strict scanning.	No
Step 1 touches	Purple EMA9 range distance <=3.00% before mandatory filters.	No
Strict candidates	Touch/near-touch rows that also pass all mandatory higher-timeframe gates.	Not by itself
Early ready	Strict rows whose early entry snapshot is entry_candidate.	Eligible during market hours
Final ready	Strict rows whose final entry snapshot is entry_candidate.	Eligible during market hours
New entry alerts	Fresh eligible entries actually persisted this run.	Yes
New exit alerts	Open IDs whose exit condition was persisted this run.	Yes

Count relationship

Step 1 rows are the widest set. Strict candidates are a subset. Entry-ready rows are a subset of strict candidates. New alerts are only fresh, eligible entry-ready events created during allowed hours.

Counts are time-sensitive

The reference numbers document one cached-data run. They will change as candles update, F&O membership changes, data becomes available, and rule parameters are refined.

13 Data requirements and lookback

Data	Why needed	Current practical window
Monthly	EMA89, purple distance, Chande Kroll, VWMA20, structure.	3000 calendar days for the Monthly profile.
Weekly	Weekly touch and Daily-profile higher confirmation.	730 days for Weekly; all-profile refresh requests 3000 days.
Daily	Daily touch, Monthly final entry, and derived higher aggregations.	365 days minimum; all-profile refresh requests 3000 days.
120 minute	Monthly early/exit and Weekly final entry.	Broker limits may constrain historical intraday depth.
30 minute	Weekly early/exit.	Enough for indicators plus lifecycle mapping.
10 minute	Daily early/final/exit.	Enough for indicators plus lifecycle mapping.

Why EMA89 can show Unknown

The Step 1 audit can run with as few as 9 touch candles, but EMA89 needs at least 89 candles on that exact timeframe. Downloading 3000 daily days does not automatically guarantee 89 valid Monthly rows if the symbol history is shorter, CSV derivation is missing, or a timeframe file was not refreshed. Strict qualification rejects Unknown EMA89.

Data quality principle

Missing or insufficient values are displayed as blockers/warnings. The scanner should not manufacture indicator values or silently treat Unknown as Pass.

14 Worked decision examples

Scenario	Observed state	Expected result
A - Step 1 only	Weekly EMA9 lies inside candle range, but close is below EMA89.	Appears in Step 1 with EMA89 blocker; absent from strict candidates; no alert.
B - Strict but waiting	Monthly purple distance <=3%; all strict gates pass; 2H close remains below yellow.	Appears in strict candidates; early=wait; final depends on Daily; no early alert.
C - Early ready	Weekly strict gates pass; Candle 2 closed; no low break; 30m close > yellow; blue > purple.	early=entry_candidate. Fresh market-hour run can create early trade ID.
D - Final blocked	Final timeframe close > yellow but yellow <= brown VWMA20.	final=wait because yellow-above-brown is mandatory.
E - Discarded	From Candle 3 onward, an entry-timeframe candle breaks the lower of Candle 1 and Candle 2 lows.	Entry status=discarded; setup cannot create an entry alert.
F - Exit	Open trade ID exists and latest configured exit-timeframe candle closes below yellow.	Exit alert is persisted and matching trade ID is closed.

Telegram delivery

When Telegram is configured and enabled, newly persisted entry and exit events can be sent to the configured chat. Candidate rows that do not create a fresh database event are not repeatedly sent. Market-hour gating and open-trade duplicate suppression reduce notification volume.

Audit sequence for an unexpected alert

Open the trade ID -> identify profile and entry kind -> inspect Step 1 range distance -> inspect each strict gate -> inspect Candle 1/Candle 2 lifecycle -> verify entry close/yellow/blue/brown -> confirm market-hour and Telegram status.

15 Review checklist for Krishna

These questions focus on places where visual chart judgment was translated into deterministic code.

Topic	Current implementation	Review decision
Purple touch	Range distance $\leq 3\%$; UI bands through 0.00%, 0.20%, 0.50%, 1.00%, 1.50%, 2.00%, and 3.00%.	Compare candidate quality by distance band.
Touch timing	Only the latest higher-timeframe candle qualifies.	Confirm whether a prior touch should remain active while lifecycle waits.
Blue approach	Latest 3 candles; prior close within 3% of blue; latest moves toward purple.	Confirm geometry and replace 3% if needed.
5-month candle	Stable calendar-derived 5-month buckets.	Confirm calendar versus rolling construction.
Higher confirmation	Latest confirmation candle only.	Confirm open/close rules and whether incomplete candles are allowed.
EMA26	Strict close $\geq$ EMA26; EMA9 > EMA26 adds context.	Confirm whether both relationships are mandatory.
Market structure	Explicit downtrend blocks.	Confirm whether this filter should remain mandatory.
C1/C2 invalidation	Any later low below the lower of the two reference lows discards setup.	Confirm wick-low versus candle-close break.
Exit	Exit-timeframe candle close below yellow.	Confirm whether re-entry is allowed after an exit and new touch.

Recommended review method

Choose 10 known charts: clear passes, clear failures, and ambiguous cases. Record Krishna's expected result for every gate, then compare the UI row field-by-field before changing thresholds.

16 Implementation map and status glossary

Primary code locations

Area	File / function
Profiles and all technical rules	trading_analysis/analysis/krishna_setup.py
Step 1	scan_krishna_purple_step1_candidate()
Strict scan	scan_krishna_purple_touch_setup()
Entry lifecycle	_purple_entry_snapshot()
Higher confirmation	_purple_higher_confirmation()
Blue approach	_approaches_purple_from_blue()
Scan orchestration	AnalysisService.scan_krishna_purple_touch_alerts()
Trade IDs and persistence	KrishnaPurpleAlertRepository in trading_analysis/storage/sqlite.py
Web presentation	web/index.html and web/app.js

Status glossary

Status	Meaning
step1_only	Purple range distance is $\leq 3\%$, but one or more mandatory strict gates failed/are unavailable.
qualified	All higher-timeframe strict gates passed; entry status is evaluated separately.
wait	Setup remains valid but entry confirmation is not yet complete.
entry_candidate	All rules for that early/final entry snapshot currently pass.
discarded	A candle from Candle 3 onward broke the lower of Candle 1 and Candle 2 lows.
missing	Required cached timeframe candles do not exist.
insufficient	Some candles exist, but not enough to calculate the required indicators.
open	Entry event has a trade ID and no matching exit yet.
closed	Matching exit event has been created.

Document maintenance

Regenerate this PDF after material rule, timeframe, status, or alert-lifecycle changes. The generator is `scripts/generate_purple_touch_guide.py` and the output is `docs/Purple_Touch_Implementation_Guide.pdf`.